machine. In the second case, we add another virtual machine to help the existing one process the extra workload.

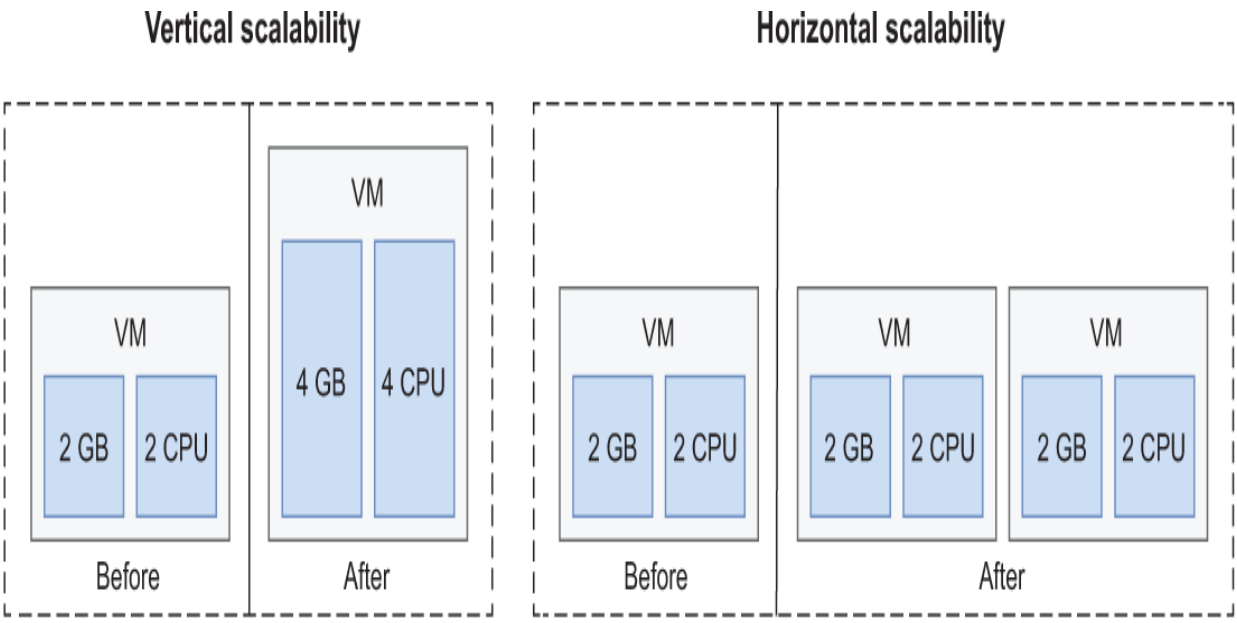


Figure 1.5 When you need to support increasing workloads, the vertical scalability model will add hardware resources to the computing node, while the horizontal scalability model will add more computing nodes.

As you'll see when we discuss Kubernetes, the platform (whether CaaS, PaaS, or something else) takes care of scaling applications in and out dynamically. As developers, it's our responsibility to design applications that can be scaled. The big obstacle to scalability is application state, which essentially is a matter of whether an application is stateful or stateless. Throughout the book, I'll cover techniques for building stateless applications and enabling them to scale without issues. Among other things, I'll show you how to push the application state from Spring to data stores like PostgreSQL and Redis.

1.3.2 Loose coupling